Number lines

1 State the integer where the point is plotted on the following number lines:

a



b



c



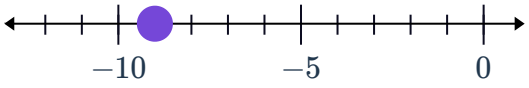
d



e



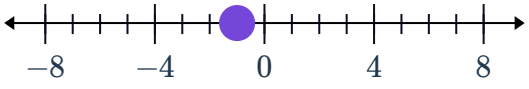
f



g



h



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i



j



2 State the number that is 6 units to the right of the plotted point:



3 State the number that is 7 units to the left of the plotted point:



4 State the number that is 6 units to the left of the plotted point:



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5 State the number that is 7 units to the right of the plotted point:



6 State the number that is 9 units to the left of the plotted point:



7 State the number that is 7 units to the right of the plotted point:



Integers

8 Describe how to tell if two integers on a number line are opposites.

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9 For each of the following number lines:

i State the number that has been plotted.

ii State the opposite of the number that has been plotted.

a



b



c



d



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10 Consider the integer 8.

a State its opposite.

b True or False: "The opposite of a number is always greater than the number itself."

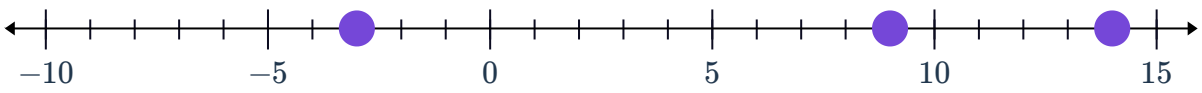
11 State the number that is the opposite of 0.

12 The opposite of a number is -16. State the original number.

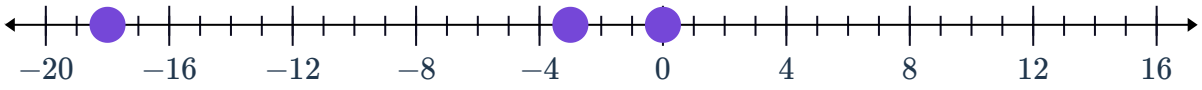
Comparison and order of integers

13 State the greatest number plotted on the following number lines:

a



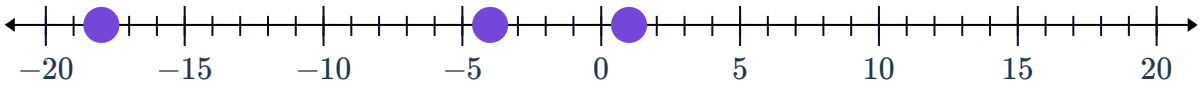
b



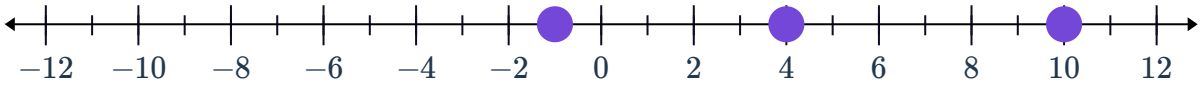
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14 State the smallest number plotted on the following number lines:

a



b



15 State the greatest number in each set:

a $13, -14, 2$ **b** $-7, -17, -12$ **c** $8, -8, 7$ **d** $35, -36, -40$

16 State the smallest number in each set:

a $-14, -4, -21$ **b** $-7, -24, 10$ **c** $1, -1, 5$ **d** $34, -36, -2$

17 Arrange the following numbers in ascending order: $10, -25, 16, -11, 27$

18 Arrange the following numbers in descending order: $3, -13, 11, -9, 15$

19 For each pair of temperatures, state the higher temperature:

a -19°C or 4°C **b** -3°C or -17°C

20 For each pair of temperatures, state the lower temperature:

a -15° C or -3° C **b** -10° C or 1° C

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21 Write an inequality that compares the two integers plotted on the following number lines:

a



b



22 For each of the following inequalities:

i Write the inequality in words. **ii** State if the inequality is true or false.

a $-7 < -2$ **b** $7 > -5$ **c** $0 > -1$ **d** $23 < -47$

23 Identify the inequality sign, $<$ or $>$, that should appear between each pair of numbers to make each statement true:

a $7 \square -4$ **b** $-9 \square 9$ **c** $0 \square -8$ **d** $-2 \square -5$

e $39 \square -72$ **f** $-12 \square -14$ **g** $-45 \square 22$ **h** $-1 \square -30$

Applications

24 Consider the following statement: "Arriving 14 minutes late."

a State the opposite of "Arriving 14 minutes late".

b Suppose "Arriving 14 minutes late" is represented by the number 14. State the integer that should represent its opposite.

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25 Consider the following statement: "Cycling $8\ \text{km}$ due south."

a State the opposite of "Cycling $8\ \text{km}$ due south".

b Suppose "Cycling $8\ \text{km}$ due south" is represented by the number 8. State the integer that should represent its opposite.

26 A gain of $\$400$ can be represented by the integer 400.

a State whether the following would represent the opposite of this situation:

i Gaining another $\$400$.

ii Losing $\$400$.

iii Losing $\$800$.

b State the integer that represents your answer in part (a).

27 State the integer that represents the opposite of the following:

a The statement "Descending 2 floors" is represented by the number -2.

b The statement "A weight gain of $11\ \text{kg}$" is represented by the number 11.

c A temperature of 10 degrees below freezing is represented by the integer -10.

28 A temperature drop of 14°C can be represented by the integer -14.

a State an integer that represents the opposite of a temperature drop of 14°C.

b State an integer that represents the opposite of the opposite of a temperature drop of 14°C.

29 A mountain peak is measured to have a height of $5100\ \text{m}$ above sea level. An ocean trench is found that is the same distance below sea level.

a If the height above sea level of the mountain peak is represented by the number 5100, state the integer that represents the location of the trench.

b In this scenario, state what the integer 0 represents.

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30 For each of the following scenarios, write down an inequality that compares the two quantities:

a Yuri has $\$8$ in savings. Kenneth has $\$13$ saved.

b Christa's account balance is $-\$10$. Ray's account balance is $-\$17$.

c Elizabeth and Roxanne both buy some tickets in the raffle at the local fete. After counting up their winnings, Elizabeth has made $\$86$, but Roxanne has lost $\$146$.

d Paul has read 27 books this year. His sister Eileen has read 17 books.

e Quiana has made $\$78.79$ this week in her part time job, and has spent $\$35.33$ on food in that time.

f The melting point of krypton is -157°C. The melting point of radon is -71°C.

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